

SAMUEL OSHIN

AI Backend Engineer · LLM Systems, RAG & Distributed Python

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PROFESSIONAL SUMMARY

AI Backend Engineer with 2+ years of experience shipping production LLM systems end to end. Sole backend engineer on a live generative AI platform: built a hybrid RAG pipeline (pgvector and full-text search in parallel) and cut LLM billing by 80% through Gemini prefix-cache anchoring. Designed a distributed regulatory-monitoring pipeline that processes 1,000+ updates daily, and ships complete client products from design to deployment. Strong in async Python (FastAPI), vector search (pgvector, Pinecone, ChromaDB), agents (A2A, MCP), and the queues, locks and ledgers that keep AI features reliable. Mentor's Choice — Best Backend Intern (HNG13); 2× HNG Finalist (top 5% of 10,000+).

TECHNICAL SKILLS

AI & LLM Systems	RAG pipelines, hybrid search (pgvector + Postgres FTS), Pinecone, ChromaDB, LangChain, Gemini, OpenAI, Anthropic, OpenRouter, prompt/prefix caching, SSE streaming, prompt engineering, semantic search
Agents & ML	A2A Protocol, MCP, tool calling, multi-model orchestration (Qwen, LLaMA, Mistral), PyTorch, Scikit-learn, fine-tuning (PEFT/LoRA)
Backend	Python 3.12+ (asyncio), FastAPI, SQLAlchemy 2.0 (sync & async), SQLModel, Pydantic, Django REST Framework, WebSockets
Distributed Systems	Celery (workers, beat), arq, RabbitMQ, Redis (distributed locks, caching), event-driven microservices, circuit breakers, dead-letter queues, idempotency
Data & Storage	PostgreSQL, MySQL, Redis, MinIO / S3, Alembic
Security & Payments	OAuth 2.0 (Google, Microsoft, Apple), JWT & token rotation, RBAC, API keys, webhook verification, Paystack, atomic transactions
DevOps & Tooling	Docker, CI/CD, Git, Pytest (asyncio), Flower, Ruff, cProfile, Playwright, Swagger/OpenAPI

PROFESSIONAL EXPERIENCE

Lead Backend Engineer (Contract) | CR8US | Remote Feb 2026 – Present

Product: CR8US Intelligence, a generative AI pre-production platform for the Nollywood film industry

- **Built the entire AI backend from scratch as sole backend engineer** using FastAPI, SQLModel and Celery, serving guest and authenticated tiers in production.
- **Kept end-to-end query latency under 1.5s** by running pgvector cosine similarity and PostgreSQL full-text search in parallel via asyncio.
- **Cut LLM token billing by 80%** with Gemini prefix-cache anchoring: dynamic, XML-structured RAG context is appended after a static system prompt so the prefix is served from cache.
- **Replaced cookie-based guest sessions** with transient in-memory state streamed over SSE, with secure guest-to-user session migration on login.
- **Hardened authentication** with dual-mode token rotation (JSON payloads and HttpOnly cookies) and 30-minute OAuth access tokens.
- **Built a credit and subscription system** on a Paystack transaction ledger that prevents unauthorized balance changes.
- **Engineered a resilient dataset ingestion pipeline** in Celery: three-stage CSV encoding detection (UTF-8-sig/CP1252), in-memory embedding reuse, and buffered bulk upserts with row-by-row transactional fallback.

Python, FastAPI, SQLModel, Celery, Redis, PostgreSQL + pgvector, Gemini, OAuth 2.0, Paystack, Docker

Backend Engineer | Emerj LLC | Remote Dec 2025 – Present

Product: Legal Watch Dog, an AI-powered regulatory monitoring platform

- **Defined the system architecture and data model** by authoring the Technical Requirements Document (TRD) and designing the ERD for multi-tenant scalability.
- **Engineered a 4-stage distributed pipeline** for real-time scraping and semantic analysis, handling 1,000+ global regulatory updates daily with FastAPI, Celery and Redis.
- **Ensured reliable multi-worker execution** with Redis distributed locks and SQLAlchemy transactional integrity (FOR UPDATE SKIP LOCKED).
- **Reduced manual document review time by ~40%** with LLM-driven semantic change detection (OpenRouter, OpenAI, Gemini).
- **Cut storage and processing costs by 30%** with SHA-256 content deduplication and high-availability MinIO storage.
- **Built scalable background processing** with dedicated Celery queues for scraping and persistence, monitored via Flower.
- **Enabled multi-tenant access** with JWT flows and multi-provider OAuth (Google, Microsoft, Apple).

Python, FastAPI, Celery, Redis, PostgreSQL, OpenAI, MinIO, Docker

Freelance Full-Stack Engineer | Selected client projects | Remote

Nov 2025 – Present

Design, frontend, backend and deployment for client products

- **Nook (e-commerce):** furniture storefront and admin on Next.js 16, Supabase Postgres and Drizzle, with oversell-safe, idempotent order transactions, server-side pricing, and Playwright end-to-end tests including WCAG 2.1 AA checks.
- **TRIS (enterprise risk auditing):** Next.js + FastAPI platform that flags risky payments with seven explainable rules, enforces separation of duties across five roles, keeps an append-only audit trail via Postgres triggers, and passes a 127-test regression suite.
- **ConnectHub (dating app):** location-aware matching on PostGIS with real-time chat, built on Next.js, FastAPI, Supabase and arq background workers.
- **Prison Gihon Global Services (real estate):** listings site with inspection booking and an admin publishing dashboard, built as a re-skinable template. **LingoMeet:** WebRTC video calls with live English–French translation via Gemini 2.5 Flash.

Backend Python Engineer Intern | HNG Internship | Remote

Oct – Dec 2025 (HNG13) · Jan – Apr 2025 (HNG12)

- **Awarded Mentor's Choice — Best Backend Intern** (HNG13); ranked in the top 5% of 10,000+ developers (HNG12).
- **Achieved zero fund loss** for the Wallet Service with atomic transactions, idempotent webhooks and Redis-backed recovery jobs.
- **Improved RAG retrieval precision by ~25%** with advanced document chunking, OpenRouter embeddings and hybrid vector search in Pinecone.
- **Scaled notifications to 1,000+/min at 99.5% delivery** with a RabbitMQ microservice using circuit breakers and dead-letter queues.
- **Reduced endpoint latency 16× (156s → 9.6s)** by refactoring to asyncio parallelism and bulk SQL updates (ON DUPLICATE KEY UPDATE).
- **Built PRRover**, an A2A-protocol agent using Google Gemini 2.0 for automated, sub-second security code reviews.

IT Support Officer (NYSC) | Coleman Technical Industries | Lagos

Aug 2024 – May 2025

- **Improved system uptime by 25%** by automating Oracle/Odoo ERP backups with Python scripts, cron scheduling and error handling.
- **Reduced downtime incidents by 20%** by resolving 100+ support tickets and implementing Group Policy controls.

Software Developer | WML-Integrated Solutions @ PZ Cussons | Lagos

Nov 2023 – Apr 2024

- **Cut IT issue resolution time by 40%** with a custom Django ticketing system featuring automated workflows and real-time notifications.
- **Prevented 15+ potential network outages** with internal Python monitoring tools for proactive alerting and logging.

KEY PROJECTS

Code Review Agent (A2A + MCP) | [GitHub](#)

- Autonomous code review agent using the A2A protocol, Google Gemini 2.0 and GitHub MCP for vulnerability detection, AST analysis and automated PR feedback in under 800ms per review cycle.

Distributed Wallet Service | [GitHub](#)

- Zero fund loss by design: atomic ledger mutations, idempotent Paystack webhooks and automated failed-transfer recovery, with JWT/Google OAuth and scoped, expiring API keys.

Project Nudge — AI Action Engine | [GitHub](#)

- Intent-extraction engine turning saved content into structured next actions with follow-up escalation, built on FastAPI, SQLAlchemy, arq + Redis workers, OpenRouter LLMs and an Expo client.

CodeBEGen — AI Backend Generator | [GitHub](#) | [PyPI](#)

- Multi-model engine (Qwen, LLaMA, Mistral) that generates production-ready FastAPI services, migrations and route handlers, streaming progress to the client in real time.

EDUCATION

B.Sc. Computer Science | Lagos State University

Feb 2019 – Feb 2024

Relevant coursework: Natural Language Processing, Distributed Systems, Deep Learning.

CERTIFICATIONS & VOLUNTEERING

- **Track Lead (IT Automation with Python)**, Mentor Me Collective: mentored 20+ participants, raising completion rates by 30%.
- **Google IT Automation with Python Professional Certificate** (2025).